

Computer Systems & Professional Practice (*Discover*)

Revision Lecture

Dr. Todd Waugh Ambridge



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Lecture Outline

- **Revision and Exams at School vs University**
- General Exam Format & Guidance
- Todd's Ten Tips (*for pwning your exams*)
- CSPP Exam Format & Guidance
- CSPP Examinable Topics
- CSPP Example Exam Questions



Revision and Exams at **School** vs University



Most of you took A-Levels, and you probably revised like this:

1. Did the first past paper
2. Checked your answers against the model solutions
3. Did the second past paper, remembering the formulaic bits from the first past paper
4. Checked your answers against the model solutions
5. Did the third past paper, remembering the formulaic bits from the first and second past papers
6. etc...



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Revision and Exams at **School** vs University



The bad news: This will no longer work on its own.

A-Level exams are based on a strict assessment criteria for all students in the UK taking that A-Level. They therefore need to be very standardised, so that they work for all pupils.

Furthermore, A-Levels focus on *recalling* key facts and standardised applications of methods to effectively the same scenario in order to test core concepts of the subject area.



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Revision and Exams at School vs **University**



Meanwhile, university exams test your *understanding* of the material and how to apply it to a variety of situations. Questions are generally (not always) less formulaic and more open-ended.

The good news: You are no longer being assessed against a strict set of standard marking criteria, but against your own ability to understand and communicate your knowledge, and ability to apply that knowledge, of the subject area.

Of course, this still means you need to memorise some things, but the focus is less on recall and more on your genuine understanding.



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Revision and Exams at School vs **University**



Another key difference is that the exam is just for your class at a particular university. It's written by your lecturers, based on the material your lecturers have taught you, and marked by your lecturers.

In my view, it's a much more genuine way of assessing your ability – the mark you receive is based on academic judgement rather than checklist-style criteria.

But: this can be daunting and tricky to adjust to. I am telling you this to be honest, not to scare you – it's true that you may need to revise a bit differently, but you can do it.



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How to do well in this module *Remember this?*

- **Attend** your lectures
 - ~~Monday 17-19 or Friday 10-12~~
(~~check your timetable!~~)
- **Use** the additional Canvas resources
 - Four recommended books
 - Bonus quizzes and exercise sheets
- **Drop-in** at our office hours
 - Feel free to ask us any questions!
- ~~**Engage** with the assignments~~
 - ~~Complete them in good time~~
 - ~~And don't waste your group's time!~~
- **Make** your own notes
 - The lecture notes are how we teach the course, but you need to learn the course in a way that suits you
- **Realise** that this advice is the same for all modules 😊



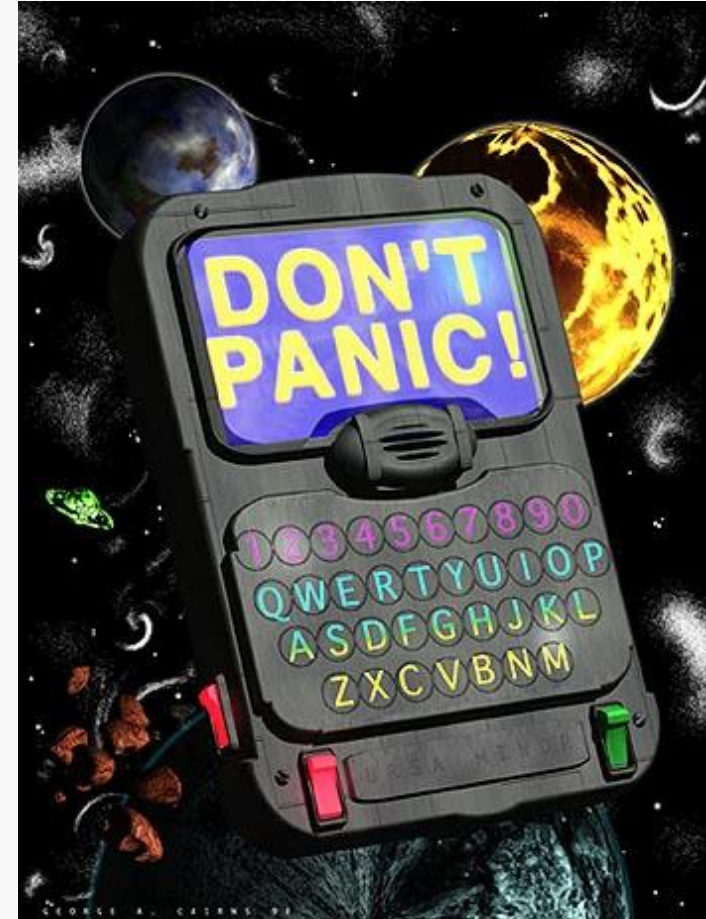
Revising at University

Half of the class are now going *“I’m so glad Dr Todd gave us this advice at the start and I’ve followed it throughout.”*

The other half didn’t follow my advice and are now panicking. **Don’t!** Although you may have a lot of work to do to prepare for the exams if you haven’t been revising as you go, it is absolutely doable and if you start now **you will be fine.**



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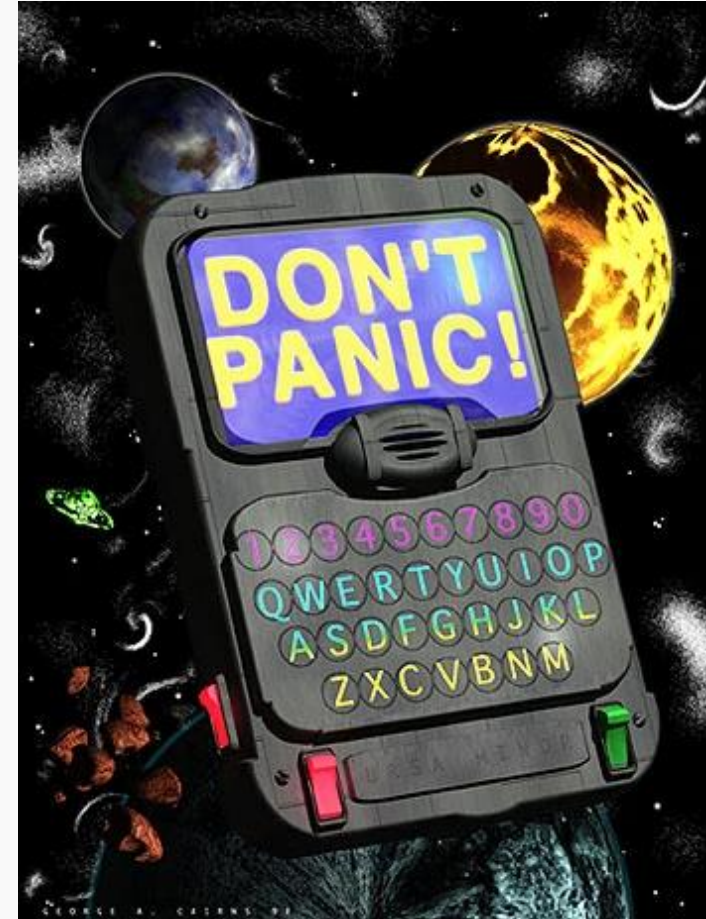


Revising at University

If you

- Revisit the lecture notes/recordings,
- Understand what *you* understand (and more importantly, what you don't)
- Utilise the additional materials,
- Make your own notes,
- Ask questions,
- And look at any past papers.

from now – you will be fine.



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When and where are my exams?



Monday 11th May – MLFCS

Thursday 14th May – AI1

Tuesday 19th May – DSA

Monday 1st June – CSPP

Wednesday 3rd June – TOC

Exams will be at various venues throughout the University campus (see your timetable!)



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What are the timings on the day?



Exams either start at 09:30 or 14:00. You are advised to arrive 20 minutes before the exam.

Exams are 2 hours long (students with RAPs may be in a different venue and may receive extra time).

If you finish early, you can leave, but you cannot leave in the first 30 minutes or last 10 minutes of the exam.



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What should I bring?



Your Student ID

Two black pens (*I strongly suggest you practice writing for a few weeks before, to avoid hand cramp – I'm serious*)

A simple scientific calculator (*except for MLFCS*)

A transparent water bottle



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What should I **not** bring?



Coats or hoodies (*leave at the back of the room*)

Your phone (*turn it off, leave under desk, leave in your bag or – if you can stomach this – don't bring it!*)

Earpods or smart watches

Calculators that can be used to store text or for programming



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What is the format of the exam?



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All of your exams have 3-4 questions, that have no overlap (*you can answer them in any order*).

Each of these 3-4 questions will have sub-questions, which will have minimal overlap (*they may share a scenario, and you may need to answer them in order – but we don't punish the same mistake twice*).

There will be a mixture of easy, medium and hard questions, as well as different question types (short answer, calculation, long answer, etc.).

Any other general advice?



Please listen carefully to the announcements at the start of the exam

This should go without saying, but obey these rules diligently

Last year, we kicked out five first-year students for various breaches of exam rules, and I'd really like to get that to 0 this year (please talk to us if you are struggling)



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Any other general advice?



Here are some University-provided resources on preparing for exams:

- [#BeExamReady](#)
- [Exam events timetable](#)
- [Managing exam stress](#)
- [Revision](#)
- [What to bring](#)
- [Exam rules](#)
- [Find your venue](#)



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Todd's Ten Tips (*for pwning your exams*)

1. Understand the course content: revisit the material and make your own notes (**don't** just “read the lecture notes”)
2. Do the past papers – but don't fall into the A-Level trap of thinking that's all you need to do
3. Finish revising early the evening before, and get a good night's sleep
4. If it's a morning exam, don't revise on the day – have a light meal instead (something to *break your fast*, as it were...)
5. Don't arrive late – in fact, arrive early!



Todd's Ten Tips (*for pwning your exams*)

6. Read the questions (**RTFQ**)! Check the *verbs* used in the question (*describe, explain, justify, calculate, state ...*)
7. Ensure you understand what the question is asking you... then make sure you actually answer it.
8. Make sure your answer is clear, understandable and legible (handwriting/English is not assessed, but you need to be able to communicate your answer clearly)
9. Think about the question and your answer from the markers' perspective. Think: "*Does my answer show understanding?*"
10. Answer **every** question!



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CSPP Exam Format & Guidance

This exam (as with all your modules except OOP) is worth 80% of your module grade.

The following topics are examinable:

1. Data Representation
2. Digital Logic Design
3. CPUs & Program Execution
4. Memory & Storage Systems
5. Processor Architecture of Modern Systems
6. Legal, Social & Ethical Considerations



CSPP Exam Format & Guidance

There will be three questions:

1. 40 marks on 2 topics
2. 40 marks on 2 topics
3. 20 marks on 1 topic

So, not all six topics will be assessed on a given paper (*no I won't tell you which one won't be – and don't try to game this*).

The resit may (or may not) assess a different set of five topics.



CSPP Exam Format & Guidance

There will be three questions:

1. 40 marks on 2 topics
2. 40 marks on 2 topics
3. 20 marks on 1 topic

Warning! This is different to the past paper. The style of questions also differs somewhat to the past paper. This is because the module team and some content has changed since last year; the past paper is still useful for revision purposes, though.



CSPP Exam Format & Guidance

There will be three questions:

1. 40 marks on 2 topics
2. 40 marks on 2 topics
3. 20 marks on 1 topic

The first two questions will be divided into a mix of short, medium and calculation questions (generally 5 or 10 marks).

The last question will be a single 20 mark long answer question.



CSPP Exam Format & Guidance

There will be three questions:

1. 40 marks on 2 topics
2. 40 marks on 2 topics
3. 20 marks on 1 topic

The questions will usually be framed by an unseen *scenario*; it is important to read this scenario carefully, so that you know what you are applying your knowledge to.

Make sure your answers consider the particular scenario!



CSPP Exam Format & Guidance

The 5-mark questions are about recalling knowledge or calculating something small.

The 10-mark questions are about applying knowledge, making design decisions which you can justify, or explaining something carefully.

The 20-mark question is about analysing a scenario and making some carefully considered decisions which you can justify clearly.

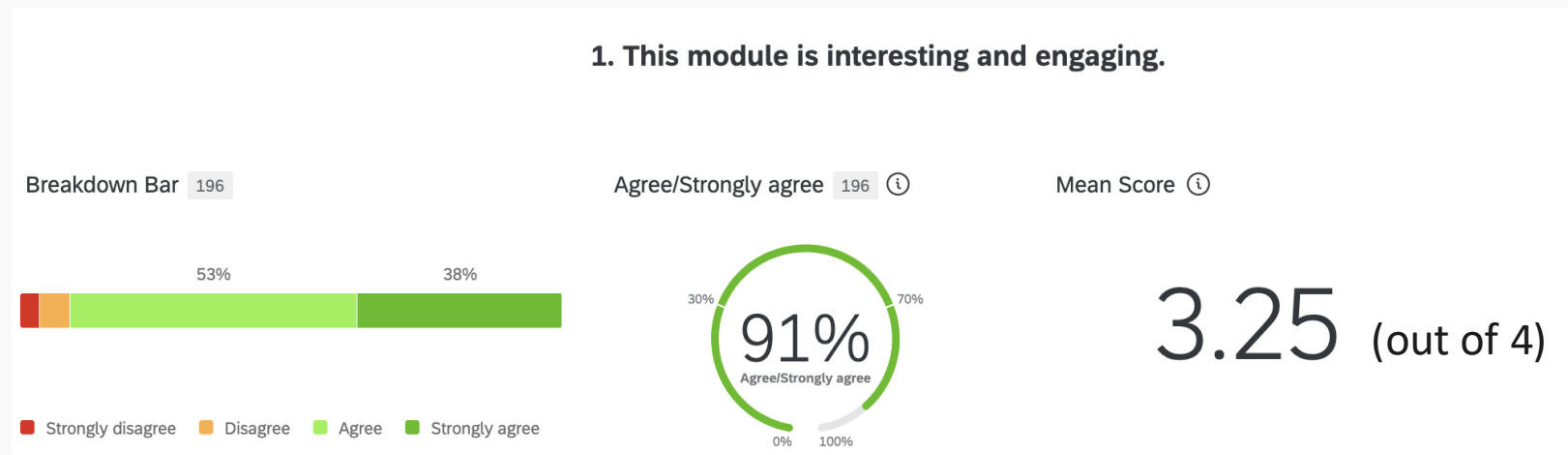


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Thank you for the feedback!



But we received one repeated criticism (that wasn't about my dress sense).

That we didn't make clear what was on the exam and what wasn't...

As I told you before, **everything is on the exam!**

Okay, fine... you win.



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Examinable Topics: **Topic 1 – Data Representation**

- The binary number system, converting between denary and binary and assessing the minimum bits required for a given task.
- Practical representations of colour (e.g. RGB) and text (e.g. ASCII).
- Representing negative numbers (sign-and-magnitude vs. two's complement).
- Representing real numbers (fixed-point vs. IEEE 754 floating-point).
- The intricacies of IEEE 754: the sign bit, the biased exponent, the hidden mantissa bit and special values (**you will be given the exponent size and bias if needed in the exam, but you will not be reminded of the other intricacies**).
- Understanding how to design representations for given practical tasks, and how to choose which representation to use for a given task, with justification.



Examinable Topics: **Topic 2 – Digital Logic Design**

- Logic circuits, truth tables and Boolean expressions; how they relate to each other and how to convert between them (e.g. using minterms/maxterms).
- Circuit design using universal gates (e.g. NAND).
- Using truth tables to verify equivalence of Boolean expressions.
- Simplification of expressions via Boolean algebra laws (**you do not need to memorise the laws, they will be given if required in the exam**).
- Karnaugh maps (including use of “Don’t Care” states); using K-maps to simplify Boolean expressions.
- Examples of combinatorial logic circuits: half/full adders, adder-subtractors, multiplexers and demultiplexers.
- Sequential logic circuits: encoders, decoders, latches and flip-flops.



Examinable Topics: **Topic 3 – CPUs & Program Execution**

- The key building blocks of computers: the CPU, ALU, CU and registers.
- The fetch–decode–execute cycle and the role of key registers (PC, MAR, MDR/MBR, IR, general-purpose registers).
- Register Transfer Language (RTL): understanding, interpreting, and classifying RTL statements into pipeline stages.
- Assembly Language: why it's used, formatting and examples of data movement, logical and branch 68008 assembler instructions (**you do not need to memorise these instructions, they will be given if required in the exam**).
- Subroutines and stacks: how return addresses are preserved; stack behaviour.
- Different addressing modes.



Examinable Topics: **Topic 4 – Memory & Storage Systems**

- The designer's dilemma for memory systems (the trade-off between low cost and high capacity and high performance), which leads to the memory hierarchy.
- Common memory components and their role in the memory hierarchy.
- Cache memory: purpose, structure, hit/miss behaviour, miss types, and why caching improves performance; as well as the limitations of caching (e.g., diminishing returns).
- Understanding of error detection and correction: noise, parity, burst errors and ECCs.
- Storage components and memory hierarchy performance motivations (when relevant to system behaviour).
- Storage technology comparisons: RAID, SSD vs HDD, access characteristics, latency, and reliability considerations.



Examinable Topics:

Topic 5 – Processor Architecture of Modern Systems

- RAM latency and the frequency wall as performance bottlenecks, and modern architectural responses (more, more, more...).
- Pipelining: structure, purpose, and how it increases throughput.
- Speculative execution: meaning, benefits, and relationship to latency hiding.
- Multicore processors.



Examinable Topics:

Topic F – Legal, Social & Ethical Considerations

- **You do not need to know the history or details of the Horizon case study, but you need to have holistic understanding of the concepts explored in that case study.**
- Identifying, escalating, and responding to faults or risks in deployed systems (both technical and procedural considerations).
- Duties related to professional responsibility, communication, governance, and prevention of harm.
- Appropriate steps to take when systems may be causing errors leading to negative legal, ethical, financial or social implications.



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Example 5-mark question (*Topic 2*)

A building-monitoring controller uses three sensors with binary outputs:

A — Fire detected,

B — Smoke detected,

C — Ventilation active.

The alarm output F is defined by the Boolean expression:

$$F = AB + \overline{A}C$$

(a) Draw a circuit diagram for the controller using only NOR gates. **[5 marks]**



Example 5-mark question (*Topic 2*)

Draw a circuit diagram for the **expression** $F = AB + \overline{A}C$ using only NOR gates. [5 marks]

How would I answer this question?

1. What do I need to represent as NOR?

- NOT
- AND
- OR



Example 5-mark question (*Topic 2*)

Draw a circuit diagram for the expression $F = AB + \bar{A}C$ using only NOR gates. [5 marks]

How would I answer this question?

2. How do I represent NOT as NOR?

- NOT A has only one input, but NOR has two
- Let's try $\text{NOT } X = X \text{ NOR } X \dots$
- When $X = 0$, $\text{NOT } X = 1$ and $X \text{ NOR } X = 1$
- When $X = 1$, $\text{NOT } X = 0$ and $X \text{ NOR } X = 0$
- Perfect! So **$\text{NOT } X = X \text{ NOR } X$**



Example 5-mark question (*Topic 2*)

Draw a circuit diagram for the **expression** $F = AB + \bar{A}C$ using only NOR gates. **[5 marks]**

How would I answer this question?

3. How do I represent AND as NOR?

- Using DeMorgan, we have $X \text{ AND } Y = (\text{NOT } X) \text{ NOR } (\text{NOT } Y)$
- We already have $\text{NOT } X = X \text{ NOR } X$
- So, **$X \text{ AND } Y = (X \text{ NOR } X) \text{ NOR } (Y \text{ NOR } Y)$**



Example 5-mark question (*Topic 2*)

Draw a circuit diagram for the **expression** $F = AB + \bar{A}C$ using only NOR gates. **[5 marks]**

How would I answer this question?

4. How do I represent OR as NOR?

- $X \text{ OR } Y = \text{NOT } (X \text{ NOR } Y)$
- And we already have $\text{NOT } X = X \text{ NOR } X$
- So, **$X \text{ OR } Y = (X \text{ NOR } Y) \text{ NOR } (X \text{ NOR } Y)$**



Example 5-mark question (*Topic 2*)

Draw a circuit diagram for the **expression** $F = AB + \overline{A}C$ using only NOR gates. **[5 marks]**

How would I answer this question?

5. We have:

- **NOT X = X NOR X**
- **X AND Y = (X NOR X) NOR (Y NOR Y)**
- **X OR Y = (X NOR Y) NOR (X NOR Y)**

The rest is just substitution!



Example 5-mark question (*Topic 2*)

Draw a circuit diagram for the expression $F = AB + \bar{A}C$ using only NOR gates. [5 marks]

$$(A.B) + (\bar{A}.C)$$
$$(A.B) + (A \downarrow A).C$$

$$((A \downarrow A) \downarrow (B \downarrow B)) + (((A \downarrow A) \downarrow (A \downarrow A)) \downarrow (C \downarrow C))$$

$$(((A \downarrow A) \downarrow (B \downarrow B)) \downarrow (((A \downarrow A) \downarrow (A \downarrow A)) \downarrow (C \downarrow C))) \downarrow (((A \downarrow A) \downarrow (B \downarrow B)) \downarrow (((A \downarrow A) \downarrow (A \downarrow A)) \downarrow (C \downarrow C)))$$

Simplify via (NOT X) NOR (NOT X) = X

$$(((A \downarrow A) \downarrow (B \downarrow B)) \downarrow (A \downarrow (C \downarrow C))) \downarrow (((A \downarrow A) \downarrow (B \downarrow B)) \downarrow ((A) \downarrow (C \downarrow C)))$$

Am I done...?



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- NOT X = X NOR X
- X AND Y = (X NOR X) NOR (Y NOR Y)
- X OR Y = (X NOR Y) NOR (X NOR Y)

Example 5-mark question (*Topic 2*)

Draw a circuit diagram for the **expression** $F = AB + \overline{A}C$ using only NOR gates. **[5 marks]**

How would I answer this question?

6. Now draw the circuit!

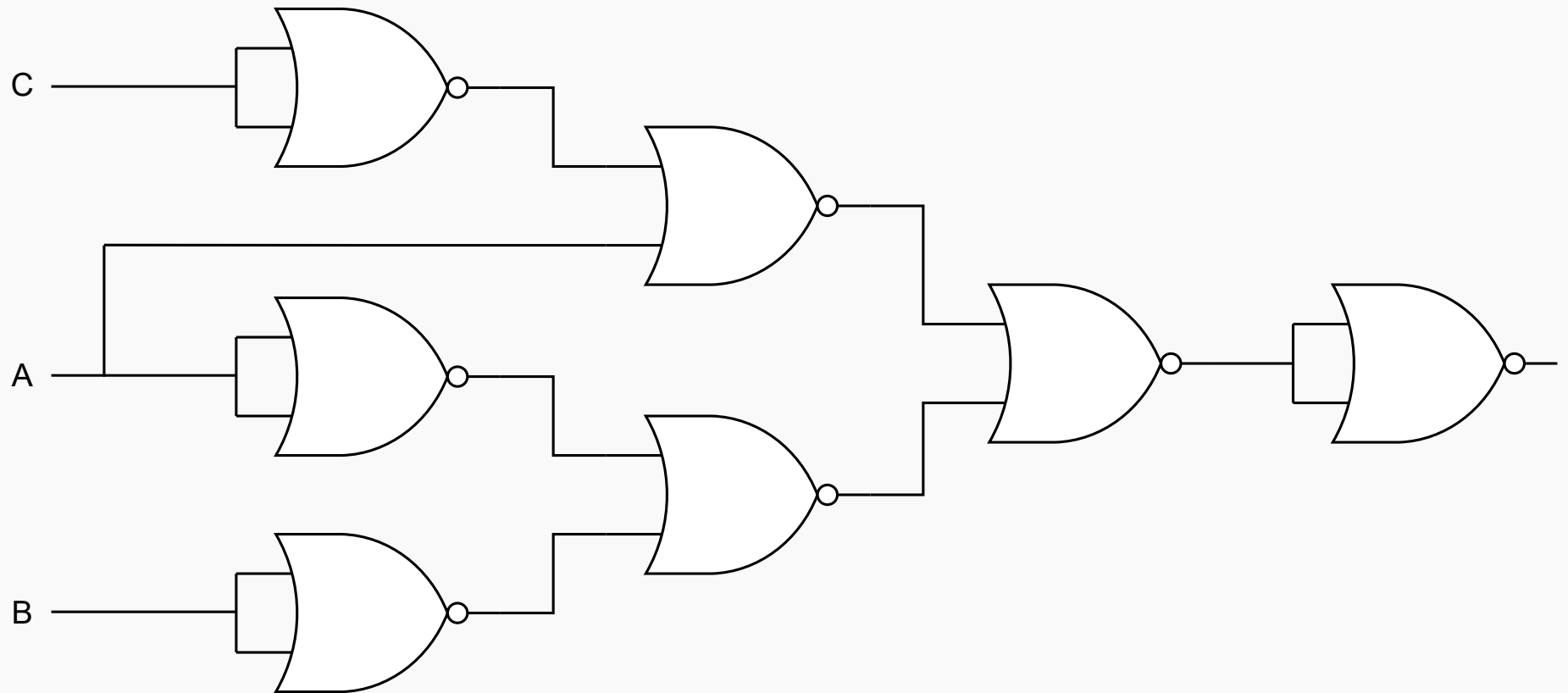


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Example 5-mark question (*Topic 2*)

Draw a circuit diagram for the **expression** $F = AB + \bar{A}C$ using only NOR gates. [5 marks]

$$(((A \downarrow A) \downarrow (B \downarrow B)) \downarrow (A \downarrow (C \downarrow C))) \downarrow (((A \downarrow A) \downarrow (B \downarrow B)) \downarrow ((A) \downarrow (C \downarrow C)))$$



Example 10-mark question (*Topic 1*)

A weather-monitoring system stores temperature readings from sensors placed across a large geographic region.

Each sensor reports temperatures in the range -60.0°C to $+45.0^{\circ}\text{C}$, recorded to one decimal place.

(a) Propose and justify a suitable representation for storing these temperature readings, and discuss one strength and one weakness of your chosen representation in this context. **[10 marks]**



Example 10-mark question (*Topic 1*)

A weather-monitoring system stores temperature readings from sensors placed across a large geographic region.

Each sensor reports temperatures in the range -60.0°C to $+45.0^{\circ}\text{C}$, recorded to one decimal place.

(a) Propose and justify a suitable representation for storing these temperature readings, and discuss one strength and one weakness of your chosen representation in this context. **[10 marks]**

RTFQ!

Four things you need to do:

1. Propose a suitable representation
2. Justify the representation
3. Discuss one strength
4. Discuss one weakness

The scenario is really important in this question.

As I said throughout the course, you need to be able to apply your knowledge to make design decisions about computer systems.



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Example 10-mark question (*Topic 1*)

My answer:

1. I would represent the temperature values as a signed fixed-point binary number with a sign bit, a 6-bit integer part and a 4-bit fractional part.
2. The sign bit is necessary to represent negative values, the 6-bit integer part can store all necessary integer values (0-60) and the 4-bit fractional part can store all ten necessary values for a significance of one decimal place.
3. One strength of this representation is that it is 11-bits, the minimum number of bits necessary to represent all 1051 possible values, as $\log_2(1051) = 10.037\dots$
4. One weakness of this representation is that hardware is generally more suited to floating point representations.



Example 20-mark question (*Topic F*)

You are working for a company that develops railway signalling systems. During testing, you discover that the software being written for an “interlocking”, which ensures railway signals show consistent indications, contains bugs that in very rare circumstances may cause a green light to be displayed when the track ahead is occupied.

What advice would you give about how this situation should be addressed? You should consider both the technical and non-technical issues this raises, including questions of safety, responsibility, and the role of regulation. Set out the advice you would give and justify your recommendations. **[20 marks]**



Example 20-mark question (*Topic F*)

This situation should be treated as a critical safety issue, not a routine software defect, because it creates a direct risk to human life. The system must not be deployed, or must be withdrawn if already in use, until the fault is fully understood and corrected.

From a technical perspective, the immediate priority is to investigate the bug thoroughly, considering logic errors, concurrency, and hardware–software interaction. After the system has been fixed, the corrected version must undergo rigorous validation and independent verification, with extensive testing.

...



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Example 20-mark question (*Topic F*)

However, non-technical issues are equally important. There is a clear ethical responsibility to prioritise safety over cost, deadlines, or reputation. The issue must be reported transparently to management, safety engineers, and relevant regulators. Concealing or downplaying the defect would be unacceptable. If any version of the system has already been used, operators must be informed immediately so that mitigating measures (such as operational restrictions) can be put in place.

In addition, the organisation must avoid overconfidence in system outputs in the future. The organisation should ensure compliance with safety standards, maintain clear lines of responsibility for decisions, and support a culture where concerns can be raised and independently reviewed.



11. If in doubt: **Ask!** First, please ask on [our Teams page](#), so that all students can benefit (and save us answering the same question 500 times!). After that, email or office hours are your best bet!

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Example 5-mark question (*Topic 3*)

Most microprocessors have a variety of addressing modes, which offer various ways for an operand to an instruction to be interpreted. As a concrete example, an instruction to LOAD something into a register with the operand “43” could do different things, depending on the addressing mode.

State three different addressing modes and provide a one sentence explanation of their function. **[5 marks]**



Example 10-mark question (*Topic 4*)

One of the main problems with modern computer architectures is RAM latency. State what RAM latency is, the effect it has on processor performance and propose one method to address RAM latency – in your answer, briefly outline this proposed method's advantages and disadvantages. **[10 marks]**



Example 20-mark question (*Topic 5*)

“Engineers should take responsibility for the consequences of the systems they design”. To what extent do you agree with this? Explain your answer.
[20 marks]

